AI Shield - Project Screenshots

Unified explainable fraud detection: scam messages, phishing URLs, emails

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AICTE | IBM SkillsBuild Machine Learning & Applied AI Internship 2026 (BharatCares)

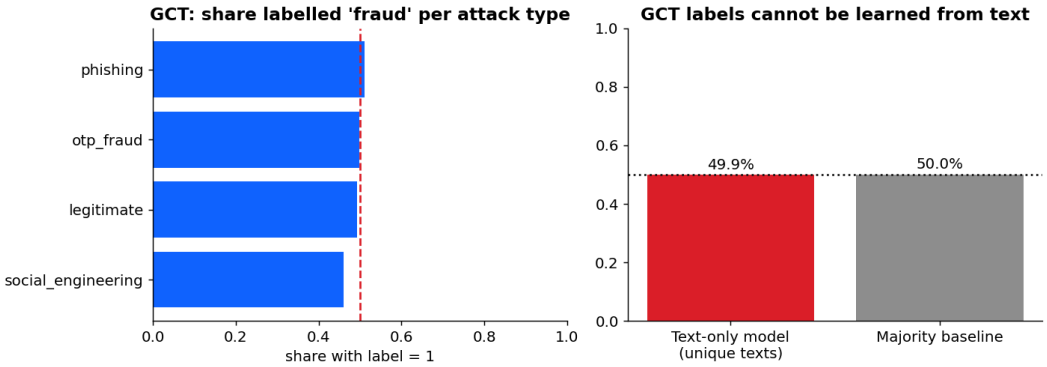
Contents: dataset audit, EDA graphs, training logs, model evaluation, predictions, tests, dashboard, architecture, app / GitHub / deployment

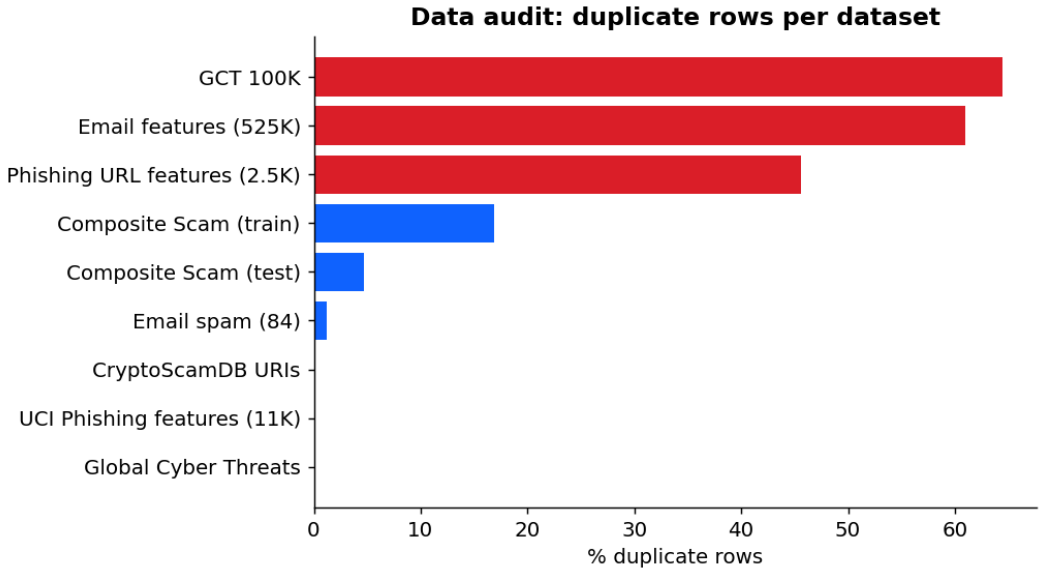
2. Dataset audit report

Dataset	Rows	Cols	Missing	Duplicates	Target	Task	Finding
Composite Scam (train)	36751	2	0.0	6205.0	label	Scam text	17% duplicate rows
Composite Scam (test)	9188	2	0.0	431.0	label	Scam text	2,900 texts leak from train
GCT 100K	100000	35	424.0	64426.0	human_verified_label	Not used	labels at chance level; 65% duplicate texts
Phishing URL features (2.5K)	2488	14	0.0	1134.0	target	URL	46% duplicates
UCI Phishing features (11K)	11055	32	0.0	0.0	Result	URL	page-content features not usable
Email features (525K)	524846	9	0.0	319794.0	label	Email (suppl.)	61% duplicate rows
Email spam (84)	84	3	0.0	1.0	type	External email test	tiny; used only for validation
Global Cyber Threats	3000	10	0.0	0.0	-	Dashboard	analytics only
CryptoScamDB URIs	17	7	2.0	0.0	-	Demo	17 rows; no labels
Enron fraud-labelled emails	447417	14	-	-	Label	Email text	0.5% fraud; corporate-only negatives

Output of eda.py -> reports/dataset_audit.csv

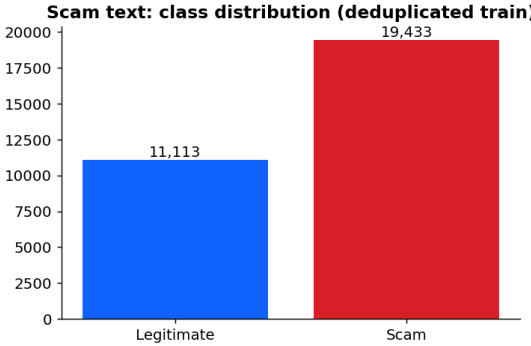
3. Data quality: labels and duplicates

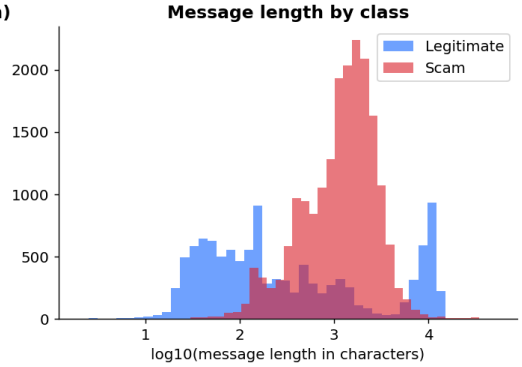


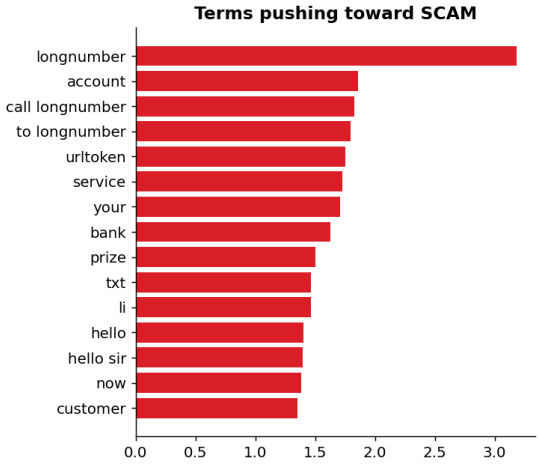


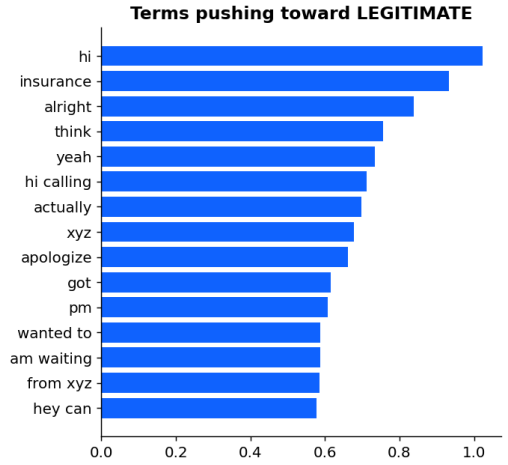
GCT labels at chance level (49.9%); duplicate rows per dataset

4. EDA: scam text



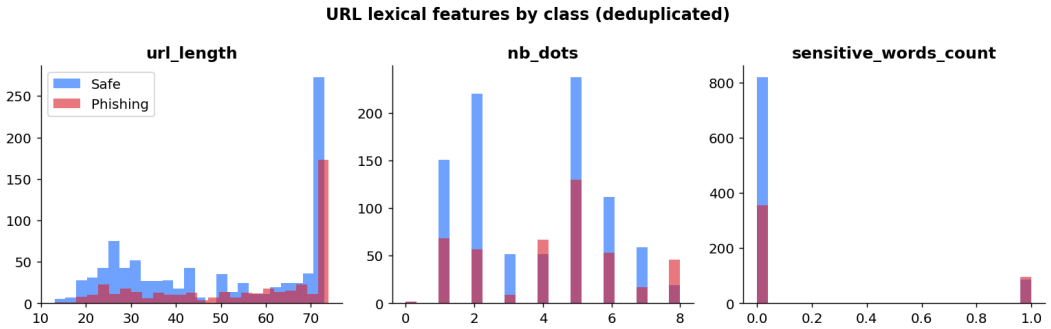


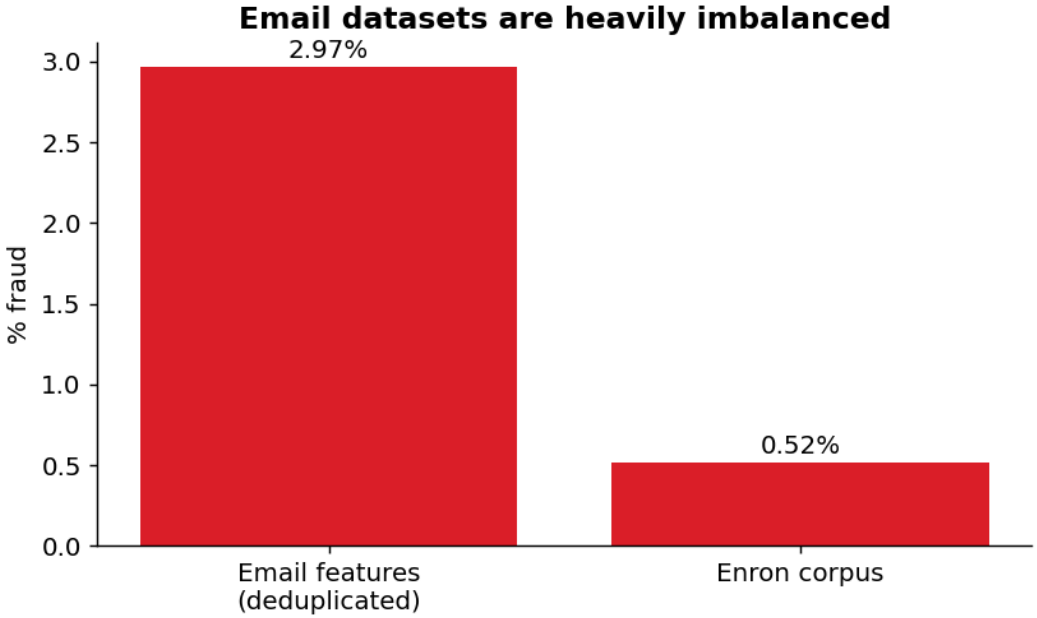




Class balance, message length, top terms

5. EDA: URL features and email imbalance





6. Training: scam-text module

```
$ python train.py   # module 1: scam text

=== MODULE 1: scam text ===
{'train_rows_raw': 36751, 'test_rows_raw': 9188, 'train_duplicates': 6205, 'test_duplicates':
431, 'test_texts_also_in_train': 2900, 'train_rows_clean': 30540, 'test_rows_clean': 5949,
'train_pos_rate': np.float64(0.6362), 'test_pos_rate': np.float64(0.7534)}
Logistic Regression 0.9937 0.9994
Naive Bayes 0.9789 0.9957
Linear SVM 0.9956 0.9995
Decision Tree 0.9802 0.9711
Random Forest 0.9903
best (validation F1, linear-text candidates): Linear SVM
TEST {'n': 5949, 'positives': 4482, 'accuracy': 0.9971, 'precision': 0.9991, 'recall': 0.9971,
'f1': 0.9981, 'roc_auc': 0.9997, 'pr_auc': 0.9999, 'confusion_matrix': [[1463, 4], [13, 4469]]}
by length: {'<200 chars (SMS-like)': {'n': 877, 'scam_share': 0.185, 'accuracy': 0.9875},
'200-1000': {'n': 1933, 'scam_share': 0.758, 'accuracy': 0.9974}, '>1000 (long dialogue)': {'n':
3139, 'scam_share': 0.909, 'accuracy': 0.9997}} | length-only baseline acc: 0.8492
```

Output of train.py (train_text)

7. Training: URL and email modules

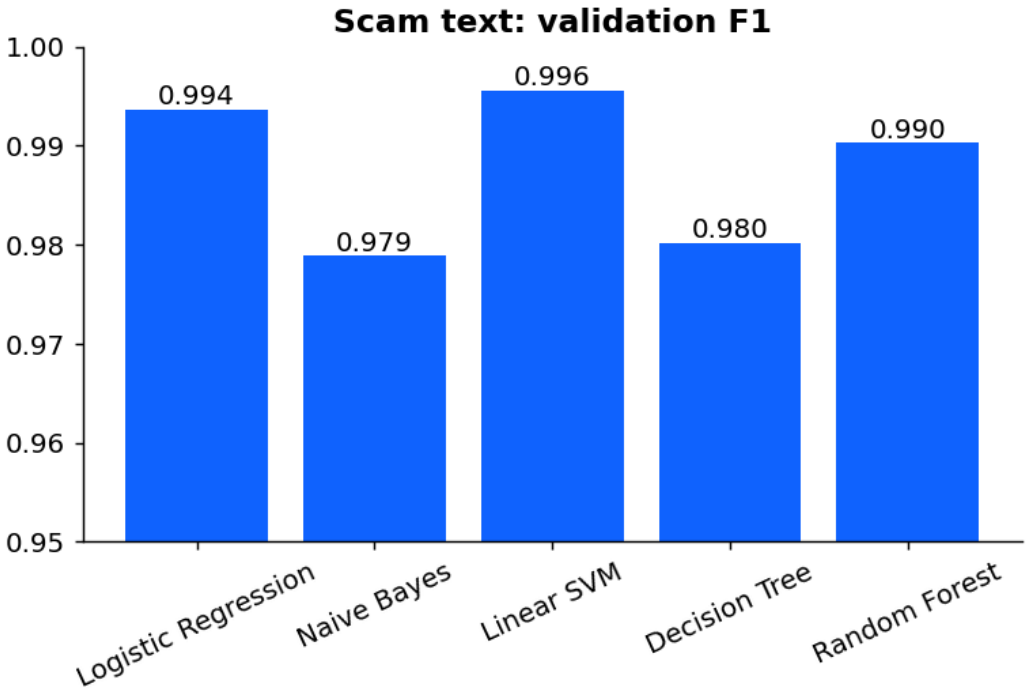

```
$ python train.py   # modules 2 and 3

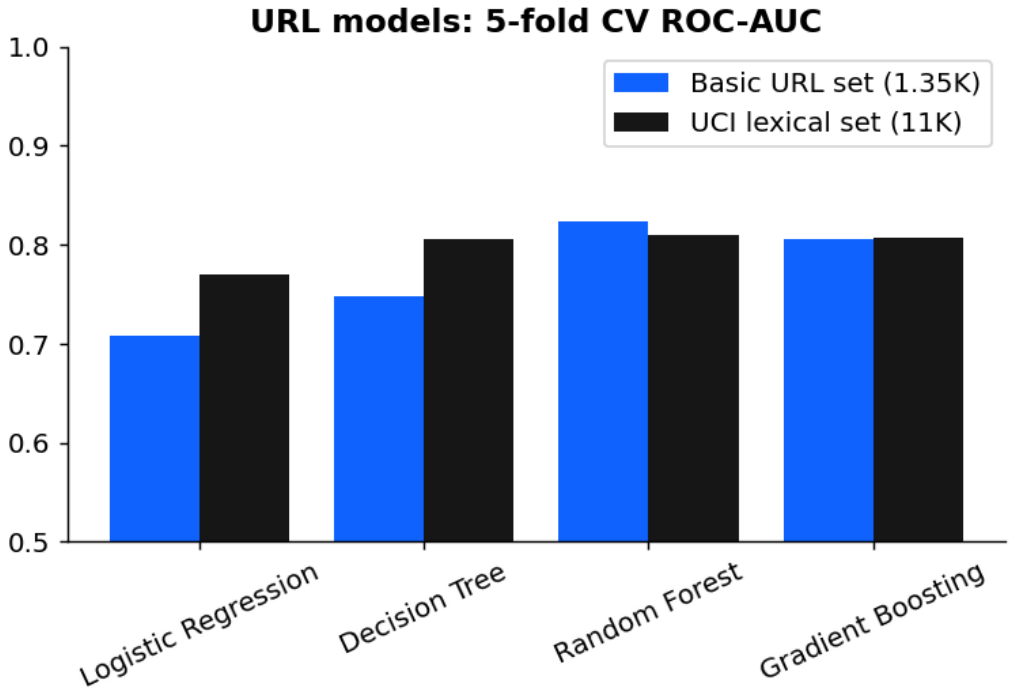
=== MODULE 2: URL ===
Logistic Regression {'cv_f1': 0.5283, 'cv_roc_auc': 0.7079}
Decision Tree {'cv_f1': 0.579, 'cv_roc_auc': 0.7477}
Random Forest {'cv_f1': 0.6337, 'cv_roc_auc': 0.8243}
Gradient Boosting {'cv_f1': 0.5836, 'cv_roc_auc': 0.8058}
Logistic Regression {'cv_f1': 0.7111, 'cv_roc_auc': 0.7701}
Decision Tree {'cv_f1': 0.7309, 'cv_roc_auc': 0.8062}
Random Forest {'cv_f1': 0.7325, 'cv_roc_auc': 0.8105}
Gradient Boosting {'cv_f1': 0.7219, 'cv_roc_auc': 0.8065}


=== MODULE 3a: email text (Enron fraud-labelled) ===
enron total 447417 fraud 2327 sampled 10451 {0: 8745, 1: 1706}
Logistic Regression 0.8836 0.9871
Naive Bayes 0.8556 0.983
Linear SVM 0.8837 0.987
external (email_spam.csv): {'n': 84, 'positives': 26, 'accuracy': 0.5476, 'precision': 0.3125,
'recall': 0.3846, 'f1': 0.3448, 'roc_auc': 0.4871, 'pr_auc': 0.3269, 'confusion_matrix': [[36,
22], [16, 10]], 'columns_used': ['text', 'type']}
```

Output of train.py (train_url, train_email_text)

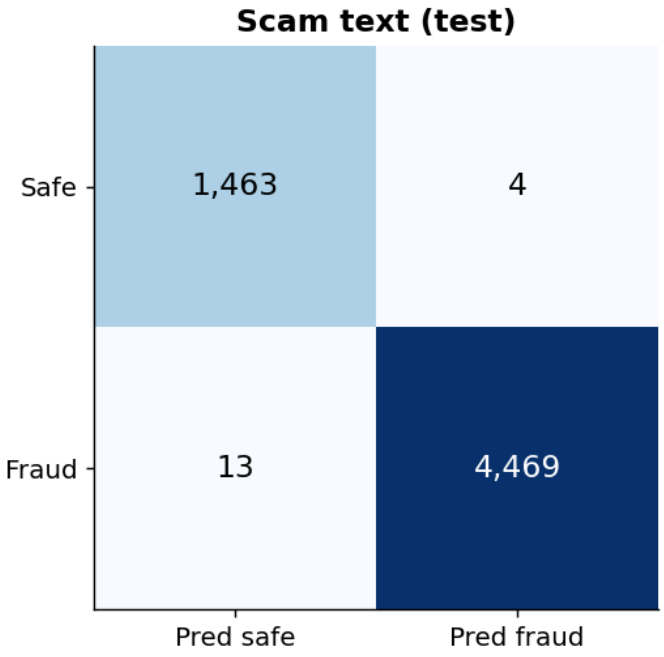
8. Model comparison

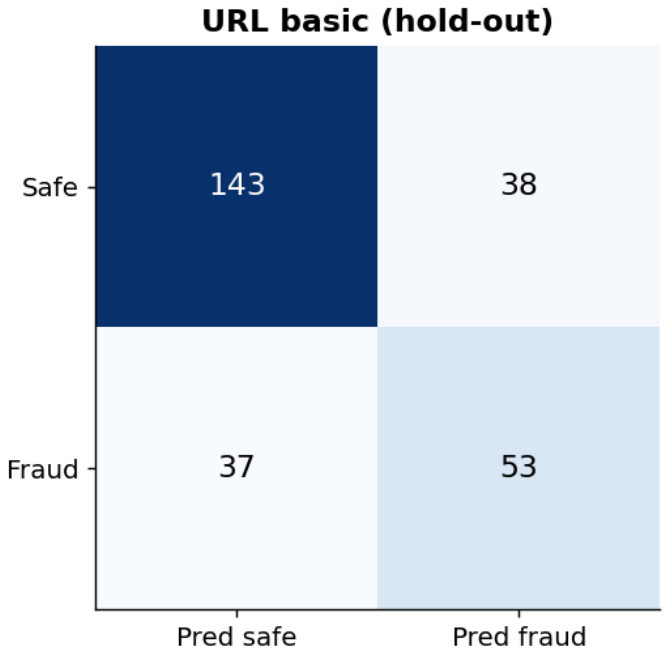


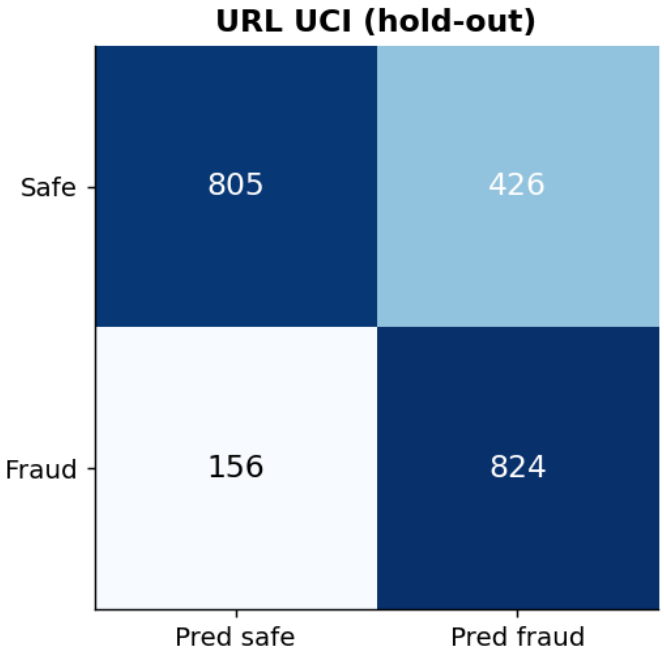


Validation F1 (text) and 5-fold CV ROC-AUC (URL)

9. Model evaluation: confusion matrices and metrics

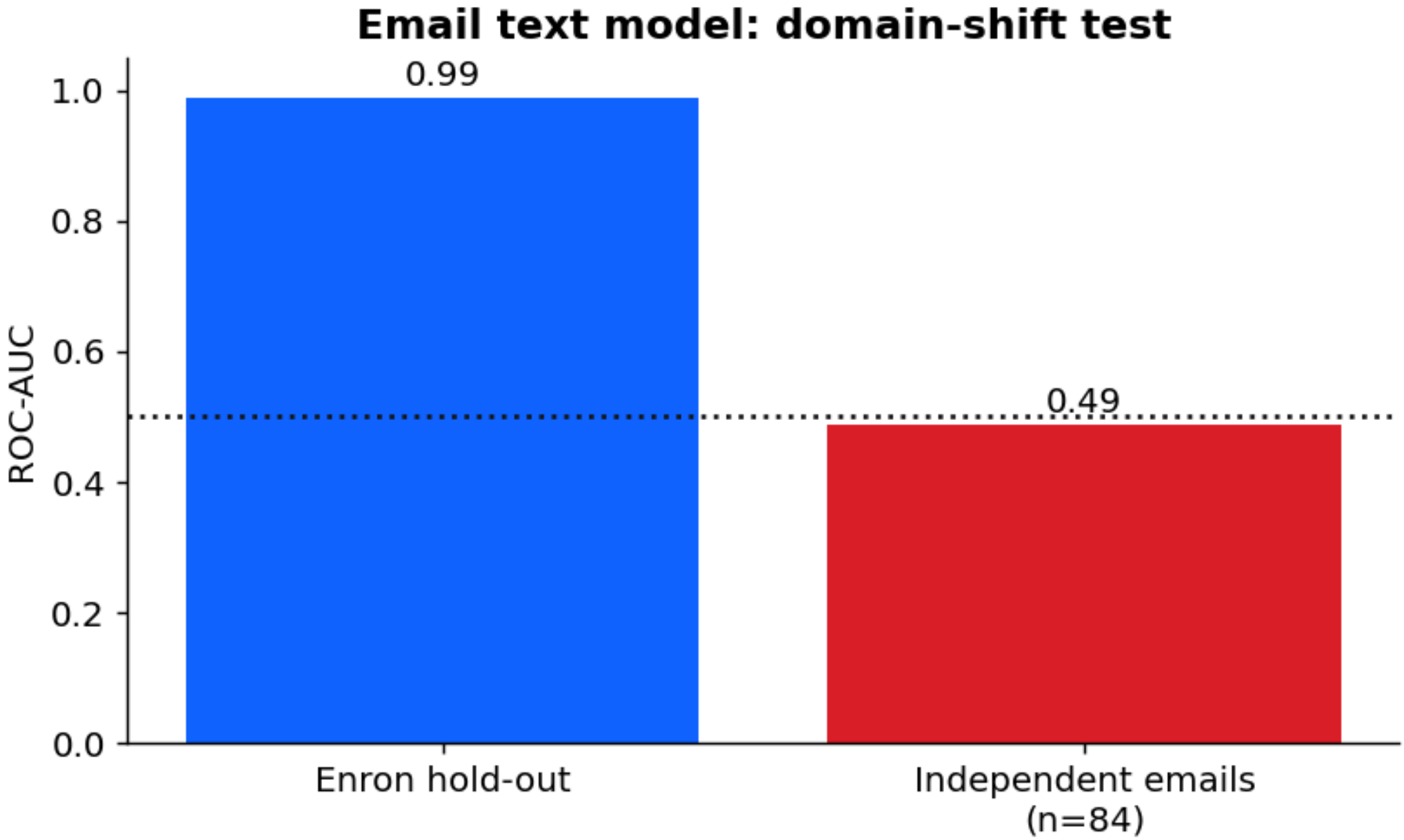






Model	n	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Scam text	5949	0.9971	0.9991	0.9971	0.9981	0.9997	0.9999
URL basic	271	0.7232	0.5824	0.5889	0.5856	0.8144	0.7312
URL UCI	2211	0.7368	0.6592	0.8408	0.739	0.8265	0.7597
Email text (Enron hold-out)	2091	0.9617	0.8761	0.8915	0.8837	0.987	0.9189
Email text (independent 84)	84	0.5476	0.3125	0.3846	0.3448	0.4871	0.3269

Held-out test data; reports/metrics.json



Held-out vs independent emails

11. Prediction output (explainable AI)


```
$ python -c 'from src.aishield.predict import ...'

TEXT: Dear customer, your SBI account will be blocked today. Update KYC immediately: http://sbi-
kyc-update.xyz/login Share OTP to verify.
  -> Fraudulent  fraud_probability=0.9898  flags=['Credential / OTP request', 'KYC / account
threat', 'Urgency', 'Authority impersonation', 'Contains a link']  terms=['urltoken', 'account',
'otp', 'kyc']

TEXT: Hey, are we still meeting at 6 for dinner?
  -> Legitimate  fraud_probability=0.0034  flags=[]  terms=[]

URL: https://www.google.com
  -> Legitimate  fraud_probability=0.0438
  red flags: []

URL: http://192.168.4.5/paytm/login.php
  -> Suspicious  fraud_probability=0.2648
  red flags: ['Uses a raw IP address instead of a domain name', 'Does not use HTTPS', 'Many
subdomains (3 dots in host)', 'Sensitive keywords: login']

URL: http://secure-paypal.com.account-verify.info/update
  -> Fraudulent  fraud_probability=0.5795
  red flags: ['Does not use HTTPS', 'Many subdomains (3 dots in host)', 'Hyphen in domain name
(common in look-alike domains)', 'Sensitive keywords: account, paypal, secure, update, verify',
'Domain ends in a TLD frequently abused for scams (.info)']
```

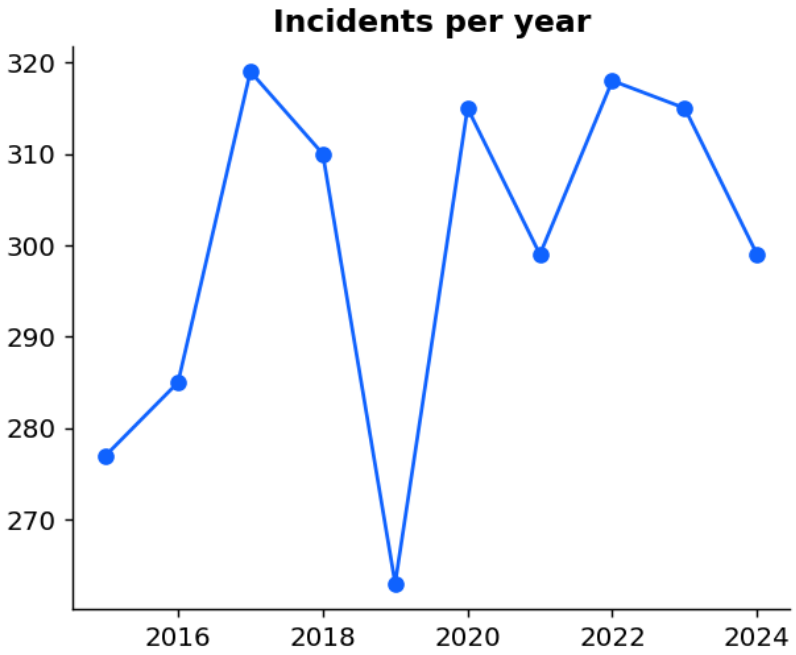
predict_text() / predict_url() from src/aishield/predict.py

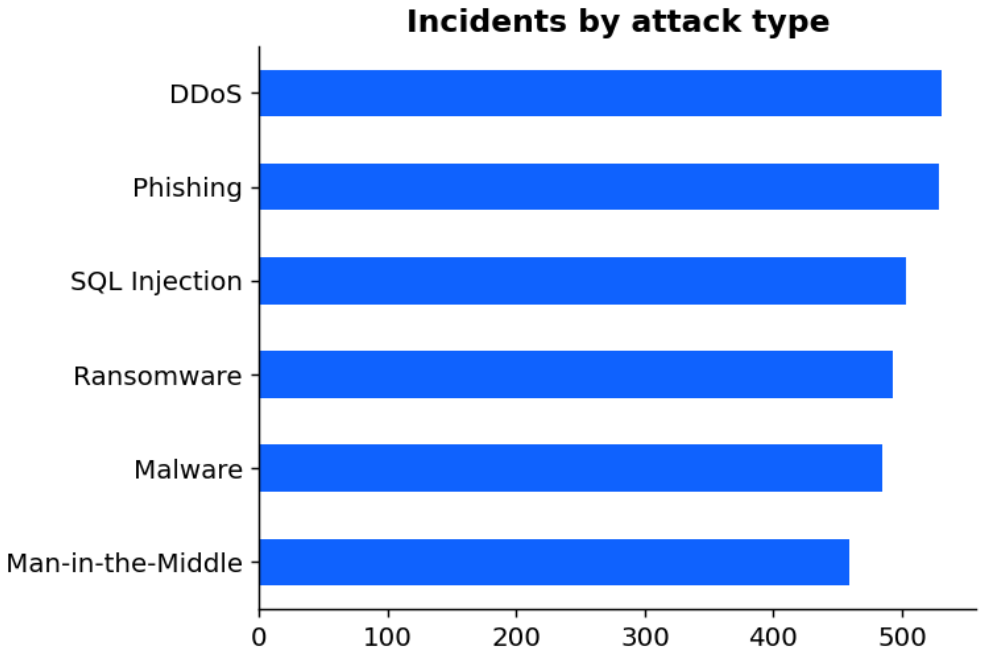
12. Unit tests

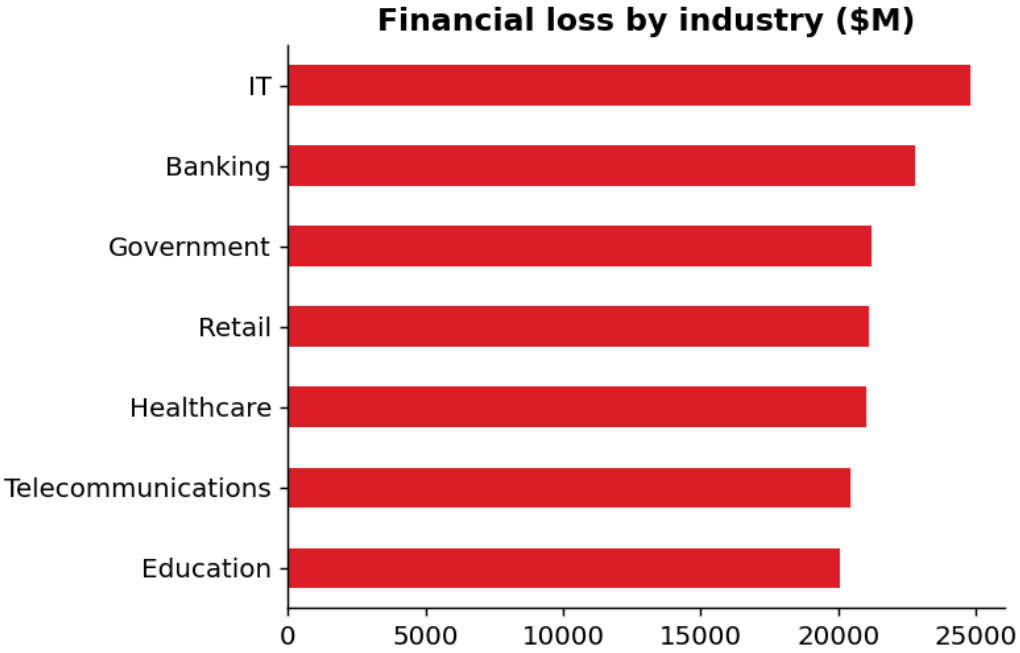
```
$ pytest-style run of tests/test_ai_shield.py
PASSED test_clean_text_masks_urls_and_html
PASSED test_url_features
PASSED test_email_structure
PASSED test_scam_and_safe_text
PASSED test_url_verdicts
PASSED test_email_returns_structure
6 tests passed
```

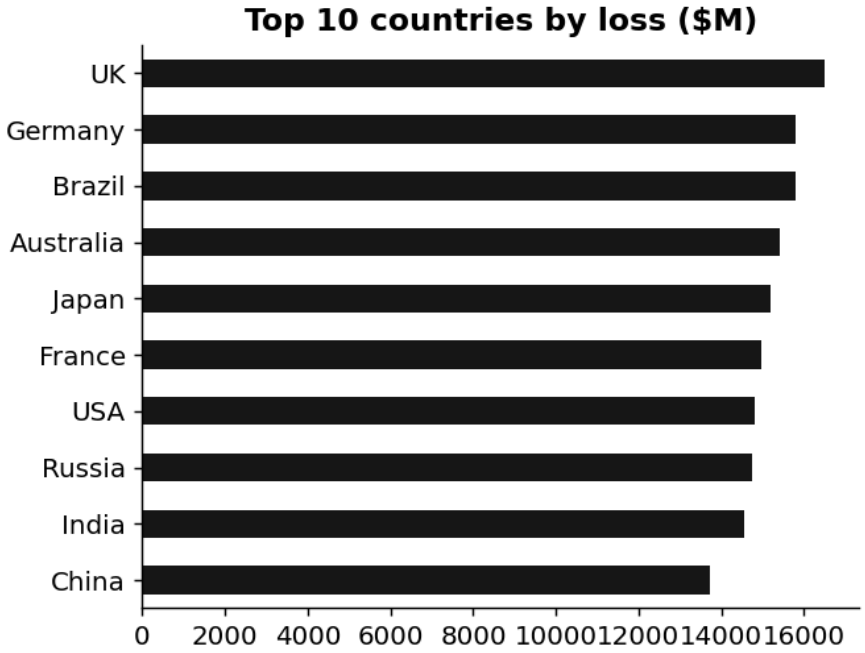
tests/test_ai_shield.py

13. Threat dashboard data (Global Cybersecurity Threats 2015-2024)

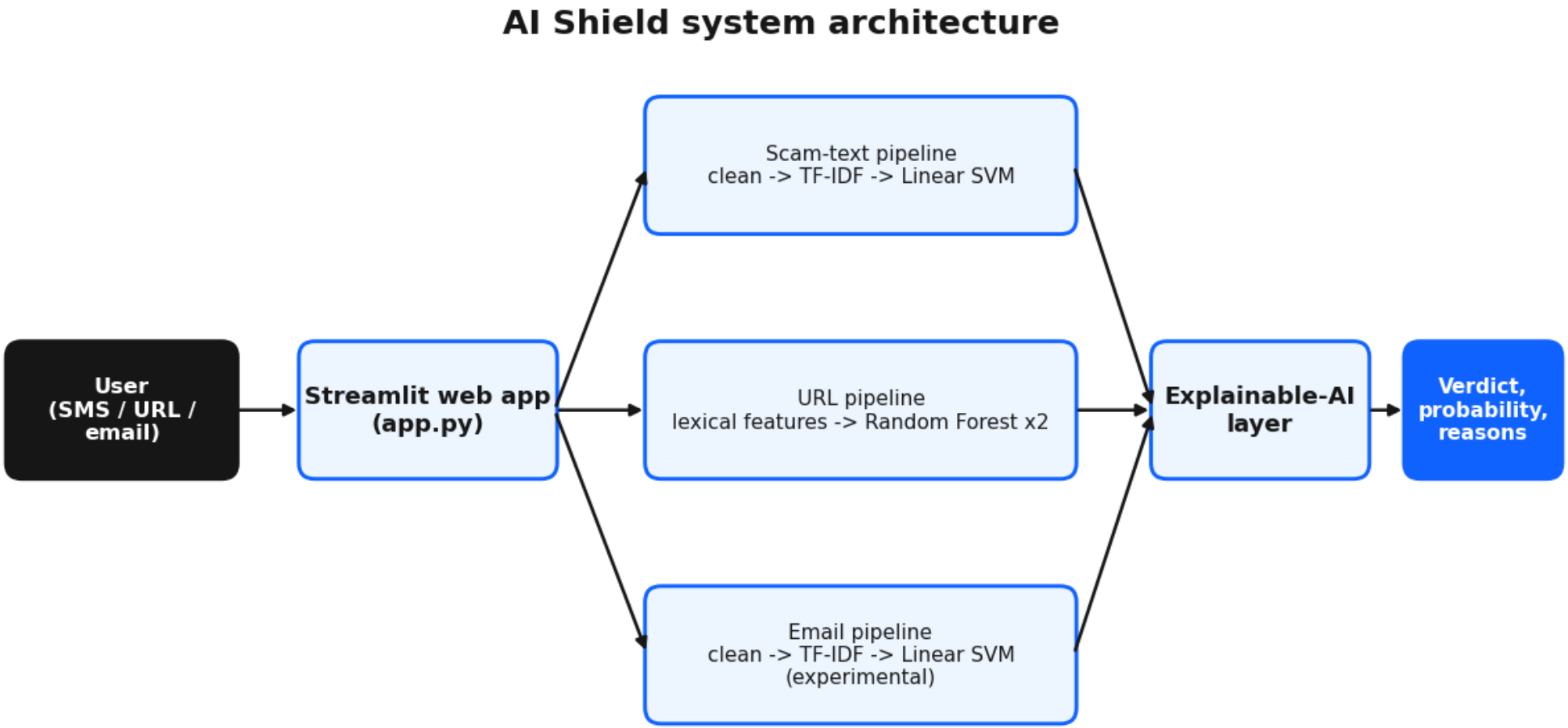








Same charts the app's dashboard page displays



Offline: raw datasets -> audit & cleaning (dedupe, leak removal) -> training & model comparison -> models/*.joblib + reports/metrics.json